

Project Report

Project Title: Import Data using Transform Maps (Spreadsheet) **Platform:** ServiceNow

Team:SWTID-2026-3488

1. INTRODUCTION

1.1 Project Overview

This project imports employee records from a spreadsheet (Excel / CSV) into a custom ServiceNow table. The file is loaded into an Import Set (staging) table, a Transform Map maps the source fields to the target table fields, and the transform is run to create or update records. The imported data is then verified in the target table.

1.2 Purpose

To automate bulk data entry, reduce errors and duplicates, and demonstrate a repeatable data-import process using Import Sets and Transform Maps.

2. IDEATION PHASE

2.1 Problem Statement

Administrators, HR data coordinators and trainees face slow, error-prone manual entry of spreadsheet data. Columns do not match target fields and no unique key is used, so rows fail or duplicates are created, leaving users frustrated and unsure of data accuracy.

2.2 Empathy Map Canvas

The administrator thinks "will every row land in the right field?" and fears duplicates and rework. Pains: time-consuming manual entry, unclear error messages, risk to production data. Gains: bulk import in minutes, a duplicate-free table, visible mapping and an audit trail.

2.3 Brainstorming

Ideas were grouped into data preparation, import set loading, transform map design, and validation & automation. Bulk import via Import Set + Transform Map, Coalesce on Employee ID and field-mapping validation ranked highest (high importance, high feasibility). Scheduled Data Sources and failure notifications were set aside as future scope.

3. REQUIREMENT ANALYSIS

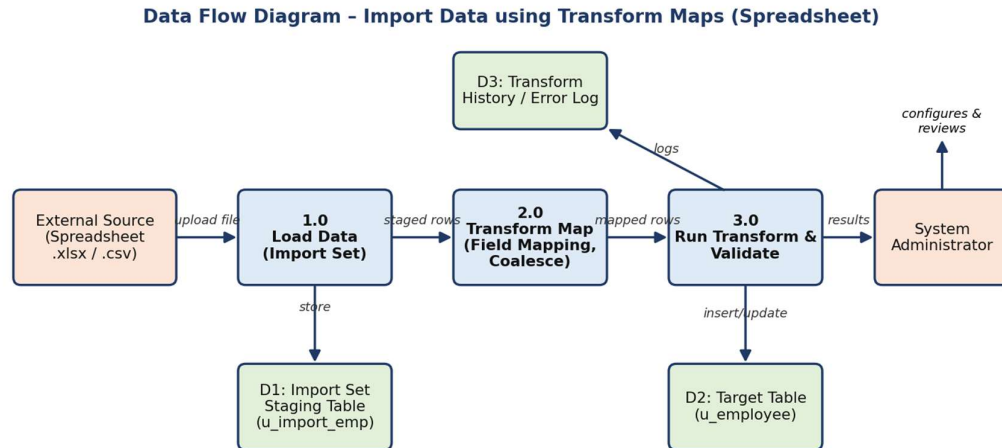
3.1 Customer Journey map

Following Entice – Enter – Engage – Exit – Extend: the admin receives the spreadsheet, cleans it and creates the target table, loads and maps the data, runs the transform and checks the results, then documents and reuses the map. Key opportunities are a fixed spreadsheet template, a pre-import checklist and error-log review.

3.2 Solution Requirement

Functional (FR-1 to FR-6): data preparation, target table setup, Import Set loading, Transform Map configuration with Coalesce, transform execution, and validation / error handling. Non-functional (NFR-1 to NFR-6): usability, role-based security, reliability, performance, availability and scalability.

3.3 Data Flow Diagram



3.4 Technology Stack

ServiceNow (Personal Developer Instance), System Import Sets (Load Data), Import Set table, Transform Map & Field Maps, Coalesce, Excel / CSV source file.

4. PROJECT DESIGN

4.1 Problem Solution Fit

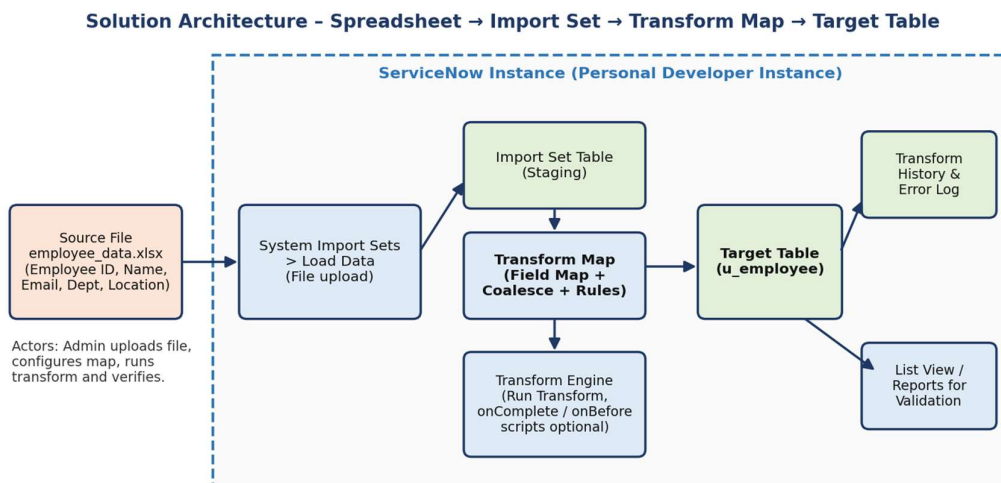
The root cause is a mismatch between spreadsheet columns and table fields, no unique key, and no staging step. The solution loads data into a staging table, maps fields with Coalesce on Employee ID, and verifies results in the Transform History, moving users from stressed and unsure to confident and in control.

4.2 Proposed Solution

A repeatable, low-code import that shows the source-to-target mapping in one place and lets data be reviewed before it reaches the target table. It saves administrator time, reduces errors, and scales to larger files or scheduled imports.

4.3 Solution Architecture

Flow: source file → Load Data → Import Set table → Transform Map (field map + Coalesce) → target table u_employee → Transform History and list view for validation.



5. PROJECT PLANNING & SCHEDULING

5.1 Project Planning

Two sprints of 6 days each, 27 story points in total, velocity 13.5 story points per sprint. [Insert the product backlog, tracker and burndown chart.]

6. FUNCTIONAL AND PERFORMANCE TESTING

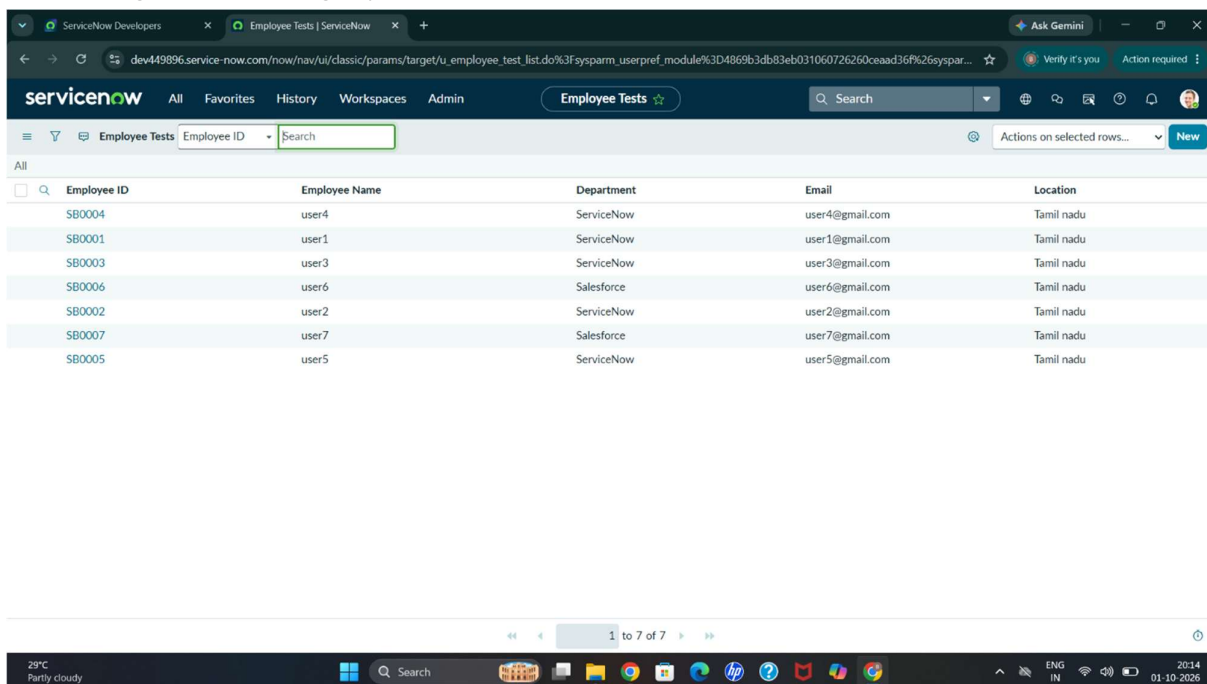
6.1 Performance Testing

8 rows were loaded into the Import Set table. 6 records were inserted and 2 records were updated. 0 records were ignored, skipped, or resulted in errors. The import accuracy was 100%, with no duplicate records created. All 5 fields (Employee ID, Name, Email, Department, and Location) were mapped correctly using Employee ID as the coalesce field. Test scenarios were executed successfully, with 6 test cases passed and 1 test case (invalid/blank data handling) not tested.

7. RESULTS

7.1 Output Screenshots

- Spreadsheet source data
- Load Data screen and Import Set table
- Transform Map and Field Maps
- Coalesce configuration
- Transform History (run result)
- Target table showing imported records



The screenshot displays the ServiceNow 'Employee Tests' interface. At the top, there's a navigation bar with 'All', 'Favorites', 'History', 'Workspaces', and 'Admin'. Below this, a search bar and a 'New' button are visible. The main area shows a table with the following columns: Employee ID, Employee Name, Department, Email, and Location. The table contains 7 rows of data, representing imported records. The bottom of the screenshot shows a Windows taskbar with various application icons and a system clock indicating 20:14 on 01-10-2026.

Employee ID	Employee Name	Department	Email	Location
SB0004	user4	ServiceNow	user4@gmail.com	Tamil nadu
SB0001	user1	ServiceNow	user1@gmail.com	Tamil nadu
SB0003	user3	ServiceNow	user3@gmail.com	Tamil nadu
SB0006	user6	Salesforce	user6@gmail.com	Tamil nadu
SB0002	user2	ServiceNow	user2@gmail.com	Tamil nadu
SB0007	user7	Salesforce	user7@gmail.com	Tamil nadu
SB0005	user5	ServiceNow	user5@gmail.com	Tamil nadu

8. ADVANTAGES & DISADVANTAGES

Advantages:

- Bulk import of records in minutes instead of manual entry
- Field mapping is visible and reusable
- Coalesce prevents duplicate records

- Staging table allows data review before it reaches the target table
- Transform History gives an audit trail

Disadvantages:

- Source data quality directly affects results; it must be cleaned first
- Initial setup of table, mapping and roles needs ServiceNow knowledge
- Large or complex transformations may need scripting
- Manual upload is required unless a scheduled Data Source is configured

9. CONCLUSION

The project shows that spreadsheet data can be imported into ServiceNow accurately and repeatably using Import Sets and Transform Maps, reducing manual effort and data errors.

10. FUTURE SCOPE

- Automate imports with scheduled Data Sources (file, JDBC or REST)
- Add transform scripts for data cleaning and reference-field lookups
- Send notifications when an import has errors
- Import related data such as departments, locations and managers

11. APPENDIX

- GitHub & Project Demo Link:
<https://drive.google.com/file/d/1NzaqnONghehdurqtCE9PuAH2YqlAK8U2/view?usp=sharing>
<https://github.com/prabha1336leo/Import-Data-using-Transform-Maps-Spreadsheet-#import-data-using-transform-maps-spreadsheet->